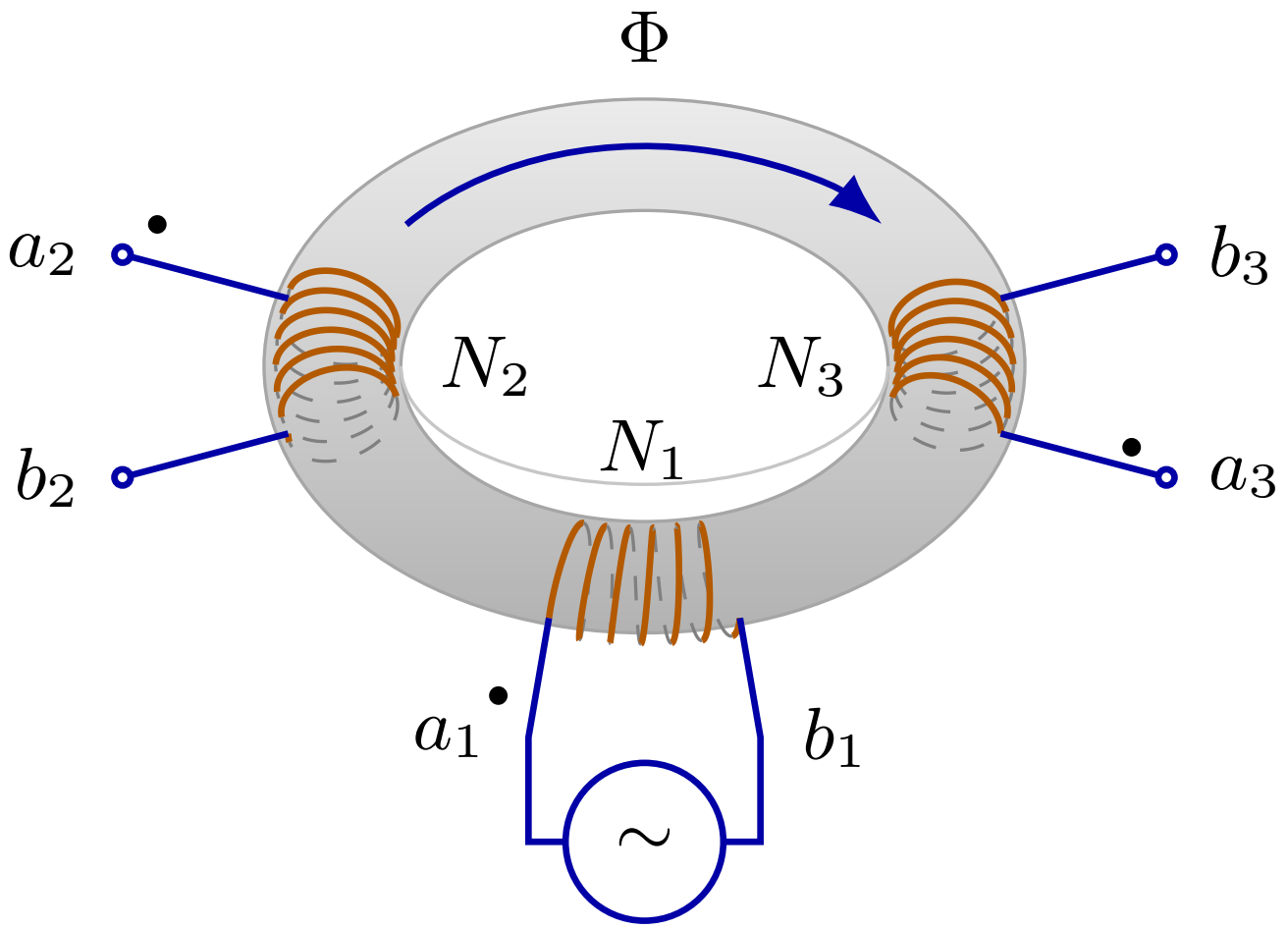

$$u_1(t) = U_0 \sin(\omega t)$$